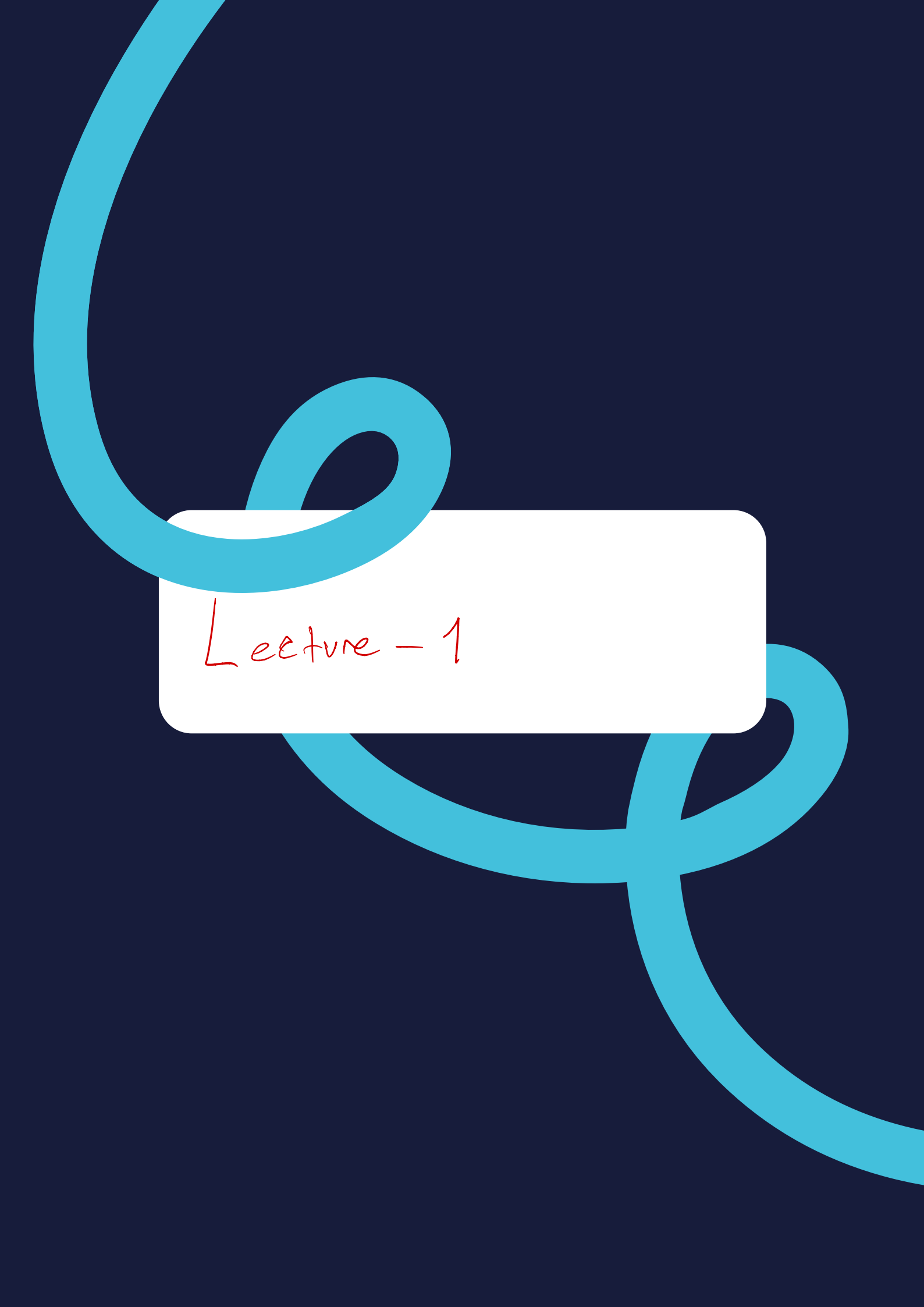




Lecture - 1

- 1- Introduction - general concepts (Dirac eq., QED)
- 2- Strong interaction
- 3 - Weak interaction

A book Modern Particle Physics - Mark Thomson (CERN Director)
Cambridge University Press

I Introduction : basic concepts

1) basics basics

Natural units

$$\hbar = \frac{h}{2\pi} = 1,055 \times 10^{-34} \text{ J.s}$$

$$c = 2,998 \times 10^8 \text{ m.s}^{-1}$$

$$1 \text{ GeV} = 1,602 \times 10^{-10} \text{ J}$$

$$\text{in Natural units} \Rightarrow \hbar = c = 1$$

Dimensional analysis

$$[\hbar] = E \cdot T = 1 \Rightarrow T = E^{-1} \text{ (Fourier Transformation)}$$

$$[c] = L \cdot T^{-1} \Rightarrow L = T \text{ (Special Relativity)}$$

NV (Natural Units) did not disappear
We need to convert back to S.I.

Example

In N.V. the Root-mean-squared charged radius of the proton is:

$$\langle R^2 \rangle^{1/2} = 4.1 \text{ GeV}^{-1}$$

We convert back to the S.I. by multiplying by $\hbar c$

$$\begin{aligned} \langle R^2 \rangle^{1/2} &= 4.1 \times \frac{1.055 \cdot 10^{-34} \times 2.998 \cdot 10^8}{1.602 \times 10^{-19}} \text{ m} \\ &= \underline{\underline{0.8 \times 10^{-15} \text{ m}}} \end{aligned}$$

Useful conversion factors

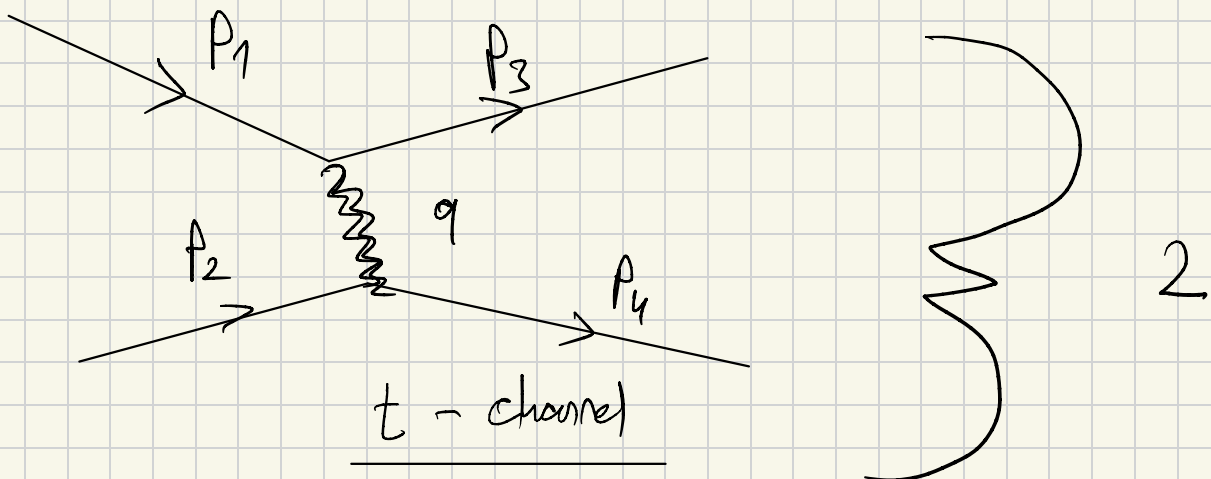
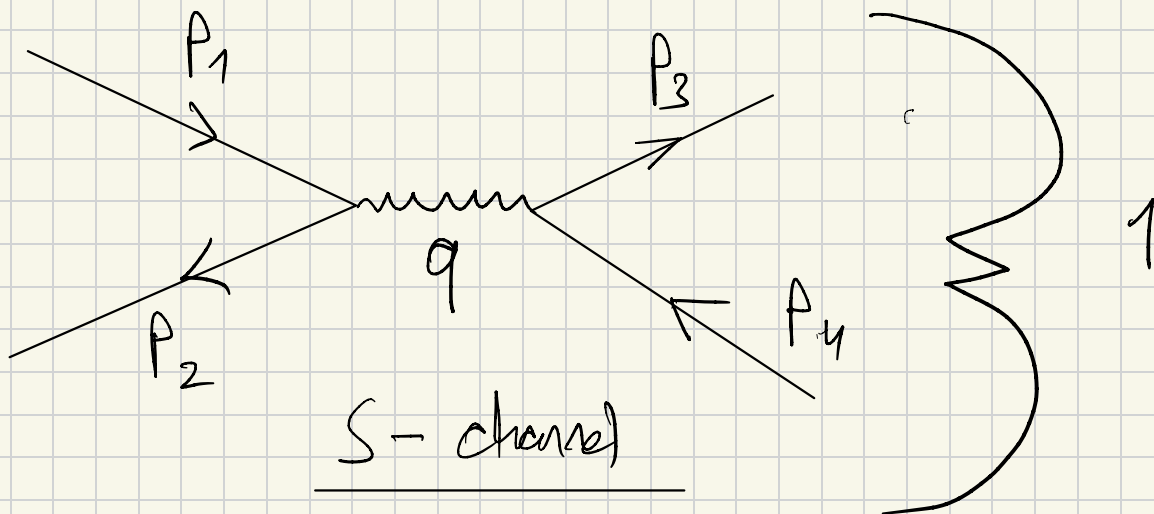
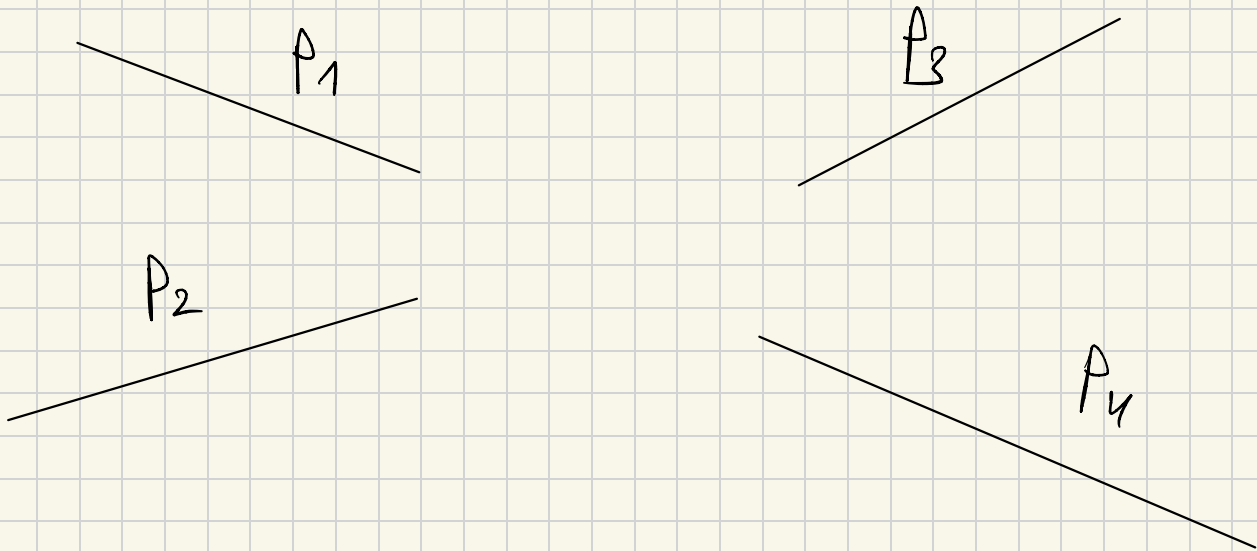
$$\hbar c = 0.197 \text{ GeV} \cdot \text{fm}$$

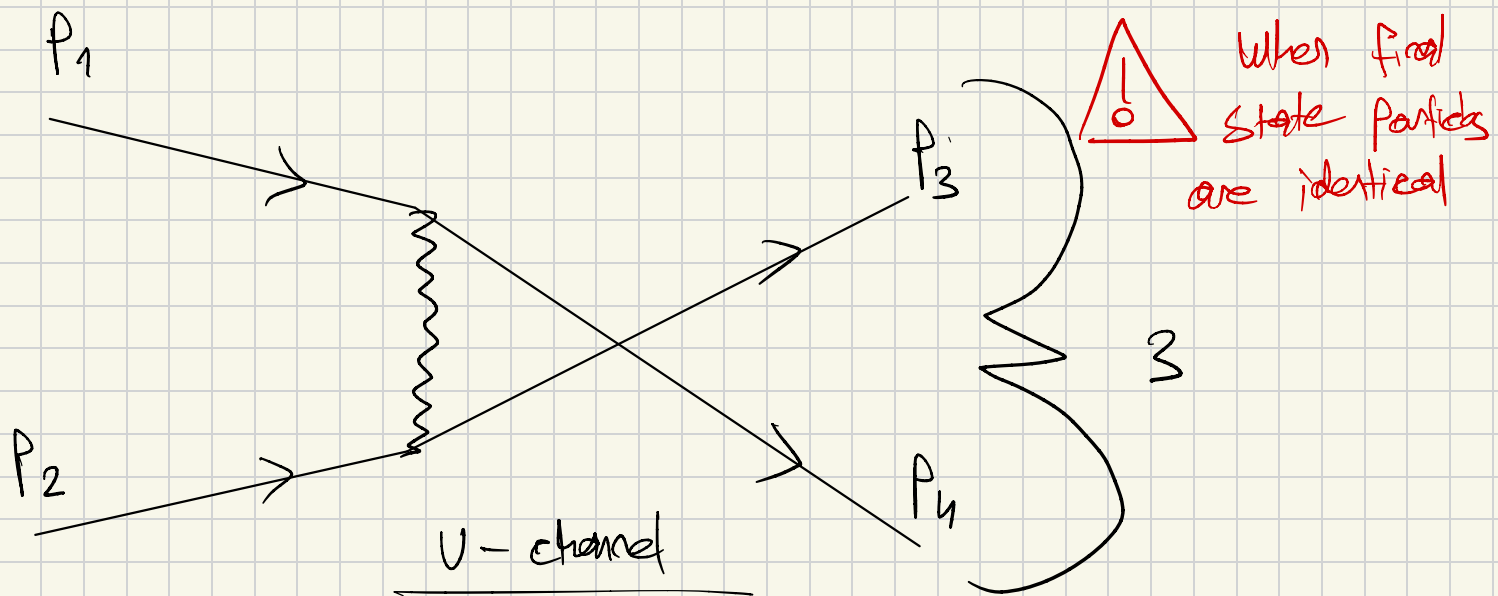
$$1 \text{ m} = 5.07 \times 10^6 \text{ eV}^{-1}$$

$$1 \text{ s} = 1.52 \times 10^{15} \text{ eV}^{-1}$$

Feynman diagrams and Mandelstam Variables

Diagrams involving the exchange of a single mediator are in 3 categories



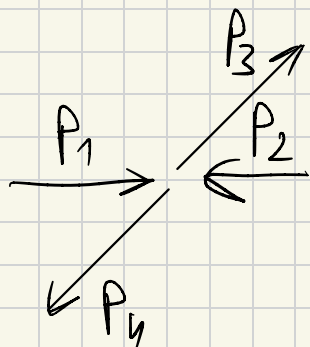


Mandelstam variables involving 2 initial states and 2 final states

$$\left. \begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2 \\ t &= (p_1 - p_3)^2 = (p_2 - p_4)^2 \\ u &= (p_1 - p_4)^2 = (p_2 - p_3)^2 \end{aligned} \right\}$$

these variables are equal to q^2 in the respective class of diagram

Case of center-of-mass of colliding particles



in COM

$$p_1 = (E_1, \vec{p})$$

$$p_2 = (E_2, -\vec{p})$$

$$s = (p_1 + p_2)^2 = (E_1 + E_2)^2 - \underbrace{(\vec{p} - \vec{p})^2}_{=0} = (E_1 + E_2)^2$$

$\sqrt{s}$: total energy in COM

2) Particle Decays (and cross-sections)

2.1) Reminders

$$N(t) = N_0 \cdot e^{-\lambda t}$$

λ : decay constant

Average lifetime $\tau = \frac{1}{\lambda}$

in Natural units: $\tau = E^{-1}$

We introduce the transition rate Γ

$$\Gamma = \frac{\hbar}{\tau} = \frac{1}{\tau} \Rightarrow [\Gamma] = E$$

2.2) QM description of decaying states

$$\psi(t) = \psi(t=0) \cdot e^{-iEt}$$

Can E be Real? No because $|\psi(t)|^2 = |\psi(t=0)|^2$

$\Rightarrow$ Yorum: Amplitude stays const. $\rightarrow$ No decay

Instead we want $E = E_0 - \frac{i}{2\tau} = E_0 - i\frac{\Gamma}{2}$

Therefore $\psi(t) = \psi(0) \cdot e^{-iE_0 t} \cdot e^{-\frac{\Gamma t}{2}}$

$$\Rightarrow |\psi(t)|^2 = |\psi(0)|^2 e^{-\Gamma t}$$

The probability density decays exponentially as expected!

To convert $\psi(t) \rightarrow \psi(E)$ we do a F.T
(See exercise session -1)

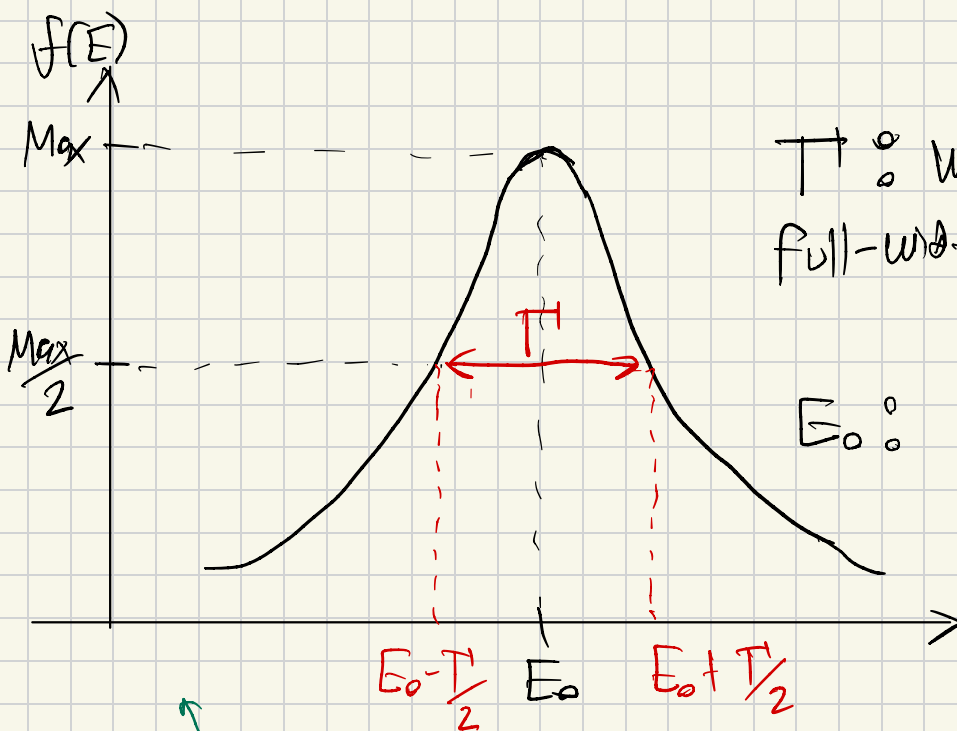
We show that

$$|\psi(t)|^2 = |\psi(E)|_{\max}^2 \times \frac{\Gamma^2/4}{(E-E_0)^2 + \Gamma^2/4}$$

Breit-Wigner Function $f(E)$

$$f(E) = \frac{\Gamma^2/4}{(E-E_0)^2 + \Gamma^2/4} = \left| \frac{\Gamma/2}{(E-E_0) + \frac{i\Gamma}{2}} \right|^2$$

Let's draw the BW function

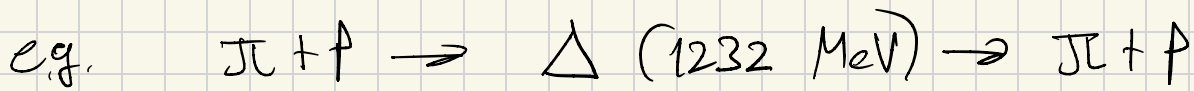
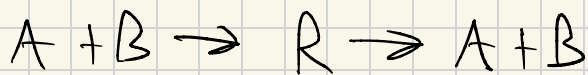


Γ : Width of the decaying particle
full-width-half-maximum of BW function

E_0 : Energy at Resonance of the particle ("on-shell")

Not : Neutron decayı bu yüzden mümkün ($n \sim 1 \text{ GeV}$ but $w \sim 80 \text{ GeV}$)
w daha düşük masslerde bulunup
decayı triggerliyor.

Example of process



$$\tau = 10^{-23} \text{ s}, \quad \Gamma = 66 \text{ MeV}$$

the cross-section of this process is proportional to $|\psi(E)|^2$

$$\sigma(E) = \sigma_{\max} \times \frac{\Gamma_{AB}^2/4}{(E - E_0)^2 + \Gamma^2/4}$$

Γ_{AB} is a partial width for $R \rightarrow AB$

Γ is the total width

Relativistic BW formula

$$|\psi|^2 \propto \frac{1}{(E - E_0)^2 + \Gamma^2/4} \Rightarrow |\psi|^2 \propto \frac{1}{(E - M)^2 + \Gamma^2/4}$$

$$\rightarrow \propto \frac{1}{\frac{(S - M^2)^2}{4M^2} + \Gamma^2/4}$$

for collisions in the COM frame

$$E_0 = M$$

$$S = E^2 - p^2 = E^2 \Rightarrow \sqrt{S} = E$$

close to the resonance $\sqrt{S} \simeq M$ so

$$E - M = \frac{(E - M)(E + M)}{E + M} \simeq \frac{E^2 - M^2}{2M}$$
$$= \frac{S - M^2}{2M}$$

$$E - M \simeq \frac{S - M^2}{2M^2}$$

$$|\psi|^2 \propto \frac{\hbar^2 M^2}{(s-M^2)^2 + M^2 \Gamma^2} \propto \frac{1}{(s-M^2)^2 + M^2 \Gamma^2} \quad \left(\begin{array}{l} \text{Relativistic} \\ \text{BW} \end{array} \right)$$

Narrow-width Approximation (NWA)

if $\Gamma \ll M$ we can use

$$\lim_{\Gamma \rightarrow 0} \frac{1}{\pi} \frac{M \Gamma}{(s-M^2)^2 + M^2 \Gamma^2} = \delta(s-M^2)$$

$$\text{Hence } \frac{1}{(s-M^2)^2 + M^2 \Gamma^2} \simeq \frac{\pi}{M \Gamma} \delta(s-M^2)$$

this allow the factorization

$$\sigma(a+b \rightarrow R \rightarrow X) = \sigma(a+b \rightarrow R) \times \left(\frac{\Gamma_X}{\Gamma} \right)$$

Branching ratio
 $B_R(R \rightarrow X) = \frac{\Gamma_X}{\Gamma}$

ex $e^+ e^- \rightarrow Z \rightarrow \mu^+ \mu^-$

$$\sigma_Z \propto \frac{\Gamma_{ee}}{\Gamma_Z} \frac{\Gamma_{\mu\mu}}{\Gamma_Z} \frac{\Gamma_Z^2}{(s-M_Z^2)^2 + M_Z^2 \Gamma_Z^2} \propto \frac{\Gamma_Z}{M_Z} \delta(s-M_Z^2)$$

Measure width of Z boson